

Yousef Kakhki

Senior Backend Engineer & Technical Lead

Tehran, Iran | Open to employer-supported relocation in Germany and the Netherlands

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SUMMARY

Hands-on senior backend engineer and technical lead with 10+ years across distributed systems, real-time media, fintech, and embedded Linux. I design system boundaries, failure handling, and operational workflows, then continue contributing directly to production code. Core experience includes Node.js, Go, Python, PostgreSQL, Kafka, NATS JetStream, WebRTC/LiveKit, Linux, and OTA delivery.

SELECTED EVIDENCE

- **5,000+ platform-level concurrent users** on an educational media platform; this figure does not claim 5,000 publishers, one room, or one SFU.
- LiveKit/WebRTC for live interactive participation, with a separate HLS playback path available hours after the session.
- Led a four-person engineering team across backend, infrastructure, and applied AI delivery.

EXPERIENCE

Solutions Architect — HonarAmoozesh

2025 · project contract

Remote

- Architected live WebRTC delivery for more than 5,000 concurrent users at platform level using LiveKit.
- Used NATS JetStream and Go for application coordination; delayed HLS playback was a separate post-session path, not a current-session fallback.

Head of Software & AI Development — Capitalino

Oct 2023 – Jun 2025

Tehran, Iran

- Led a four-person team delivering backend systems, a crypto-to-fiat payment gateway, and an AI hologram installation.
- Designed idempotent payment workflows and event-driven reconciliation patterns with explicit financial-integrity boundaries; public production and regulatory status are not claimed here.
- Led the AI hologram project from concept through delivery; the installation won Best Booth at ITEX 2024.

Senior Backend Engineer — Avin Avisia

Apr 2021 – Sep 2023

Tehran, Iran

- Built backend workflows for trading and transactional systems with Node.js, PostgreSQL, Redis, Web3.js, and messaging infrastructure.
- Worked on matching logic, order-book recovery, and transactional ledger semantics with explicit consistency and failure boundaries.

EXPERIENCE (CONTINUED)

Systems Engineer — Embedded Linux, Batna

Sep 2019 – Mar 2021

Tehran, Iran

- Designed atomic OTA update workflows with A/B partitions, rollback safeguards, and secure delivery.
- Modified and developed Yocto/Buildroot Linux distributions, firmware delivery, and kernel, boot, and release-pipeline integration.

Android Systems Developer — Azma Data Structure

Apr 2016 – Aug 2019

Mashhad, Iran

- Developed native Android modules with C++/NDK and offline-first synchronization workflows.
- Worked on local storage, background queues, conflict handling, and device-facing application delivery.

TECHNICAL FOCUS

Backend: Node.js, Go, Python, TypeScript, REST, GraphQL, WebSocket, gRPC

Distributed systems: Kafka, NATS JetStream, event-driven services, idempotency, CQRS, PostgreSQL, Redis

Real-time media: WebRTC, LiveKit SFU, delayed HLS playback, role transitions, signaling boundaries

Platform and edge: Linux, Linux Kernel, Yocto, Buildroot, ARM, OTA, Docker, Docker Swarm, Prometheus

Applied systems: Web3.js, payment workflows, ledger semantics, computer-vision integration, Android NDK

EDUCATION

M.Sc. in Computer Science (System Design)

2018 – 2021

Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran | GPA 3.8/4.0

B.Sc. in Computer Science

2012 – 2017

University of Bojnurd, Bojnurd, Iran

SELECTED TRAINING AND CREDENTIALS

IELTS 7.5 (C1) | TensorFlow for Artificial Intelligence, Coursera

Research-level study: Hyperledger Sawtooth Architecture

LANGUAGES AND WORKING MODEL

Persian: Native | English: Professional proficiency (IELTS 7.5)

Open to employer-supported relocation in Germany and the Netherlands, plus selected remote engineering or consulting engagements.

Selected public facts are evidence-bounded; proprietary deployment details and unsupported performance metrics are intentionally omitted.

Portfolio: kakhki.me | Résumé URL: kakhki.me/files/yousef-kakhki-resume.pdf